

Matteo Paolini

Mechanical engineer · mechanical design,
simulation & robot learning

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Available from February 2027

EXPERIENCE

Research Intern — Physical AI for robotic manipulation · Royal Military Academy, Brussels 2026 – present

- Building, as an independent project, a MuJoCo + reinforcement-learning pipeline that decides which object a robot should remove next from a cluttered pile to reach a buried target without collapsing it.
- Observe → decide → execute architecture: simulated stereo / RGB-D / LiDAR, segmentation detector trained on domain-randomised images, **teacher-student PPO** (privileged teacher, sensor-only student) and a gripper-reachability check.
- Training infrastructure: bank of pre-settled scenes, **16 parallel MuJoCo workers**, bit-exact resumable runs and an ablation benchmark against rule-based baselines.
- Found simulation artefacts that invalidated measurements and fixed them (settle-to-rest, rolling friction: buried-target scenes from 0/10 to 6/10). Next: simulated UR5, then sim-to-real on the physical arm.

Assembly & Machine Calibration Intern · Co.Me.Ta. Srl, Montegranaro (IT) Summer 2019

Assembled frames and integrated sensors, motors and electromechanical parts into industrial machines; final functional calibration.

EDUCATION

M.Sc. Mechanical Engineering — Mechatronics · UNIVPM, Ancona · Erasmus at ULB & VUB, Brussels (2025–26) 2024 – present

Robotics, control system design, industrial automation, machine learning, composite structures, manufacturing.

B.Sc. Mechanical Engineering · UNIVPM, Ancona 2021 – 2024

Thesis: collision-free motion planning for the redundant TIAGo arm — implemented and compared RRT and RRT*.

SELECTED PROJECTS

Soft tentacle gripper · VUB Brussels · team of 5, supervised by Prof. B. Vanderborght 2025

- Owned the tentacle: chamber geometry in ANSYS with a 2nd-order **Ogden hyperelastic** silicone model; two-part 3D-printed casting mold with removable core.
- Integrated by the team on a 2-DOF arm (belt reductions sized from a 2.40 N·m torque requirement, Arduino pneumatic control); public build guide and video.

Dual-arm object handover · Robotics II, VUB · academic project 2025 – 26

- Two Franka Panda arms pass a cube mid-air in Drake: offline handover-pose search from the arms' workspace intersection, IK, OMPL only where collisions are likely, 13-state coordinator.
- Full collision-free pick → handover → place in ≈ 20 s.

Apple Watch case — production study · UNIVPM · team of 2 2025

- Reconstructed the AI 7005-T6 case in Siemens NX with functional GD&T (± 0.01 mm fits, $\varnothing 0.05$ concentricity on the $\varnothing 3$ H7 crown bore); full process route from laser blanking to tumbling.
- Sized a **1,000,000 parts/year** line (1 laser, 89 CNC centres, 1 tumbler) from MRR and cutting power; removed a milling bottleneck by scheduling the fleet over four periods; Taylor tool-life and cost-per-part model.

Re-testing a published bearing-fault classifier · Electrical machines & drives, UNIVPM · academic project 2026

- Faithfully replicated a DAE + CNN method (Sensors, 2021) on Paderborn motor-current data: **46.9%** accuracy vs the reported 99.6%. Traced the cause to SELU output clipping (25.4% of samples unreachable); scaling the input raised AUC from 0.60 to 0.77, while other training changes had no effect.

Fault diagnosis on a multiphase plant · Smart Factories, UNIVPM · academic project 2026

- Showed a global detector learns the operating point, not the fault; with leakage-free temporal splits a regime-conditioned Random Forest detects 502–515 of 523 faults with **0 false alarms** on 798 healthy samples.

Also: vacUUUm cordless vacuum (Siemens NX product design) · 1U CubeSat CFRP/aluminium structure (ANSYS ACP) · LPBF 316L metallography and Vickers testing · PLC elevator controller (GRAFSET) · wheelchair-accessible routing (A*, MDP).

SKILLS

Robotics & ML Python · MuJoCo · Gymnasium · Stable-Baselines3 · Drake · OMPL · scikit-learn · pytest · Git

Mechanical Siemens NX · SolidWorks · ANSYS Workbench / ACP · hyperelastic FEA · GD&T & drawings · DFM · mold design · process planning

Prototyping FDM 3D printing · silicone casting · Arduino · MATLAB / Simulink

Languages Italian (native) · English (C1)